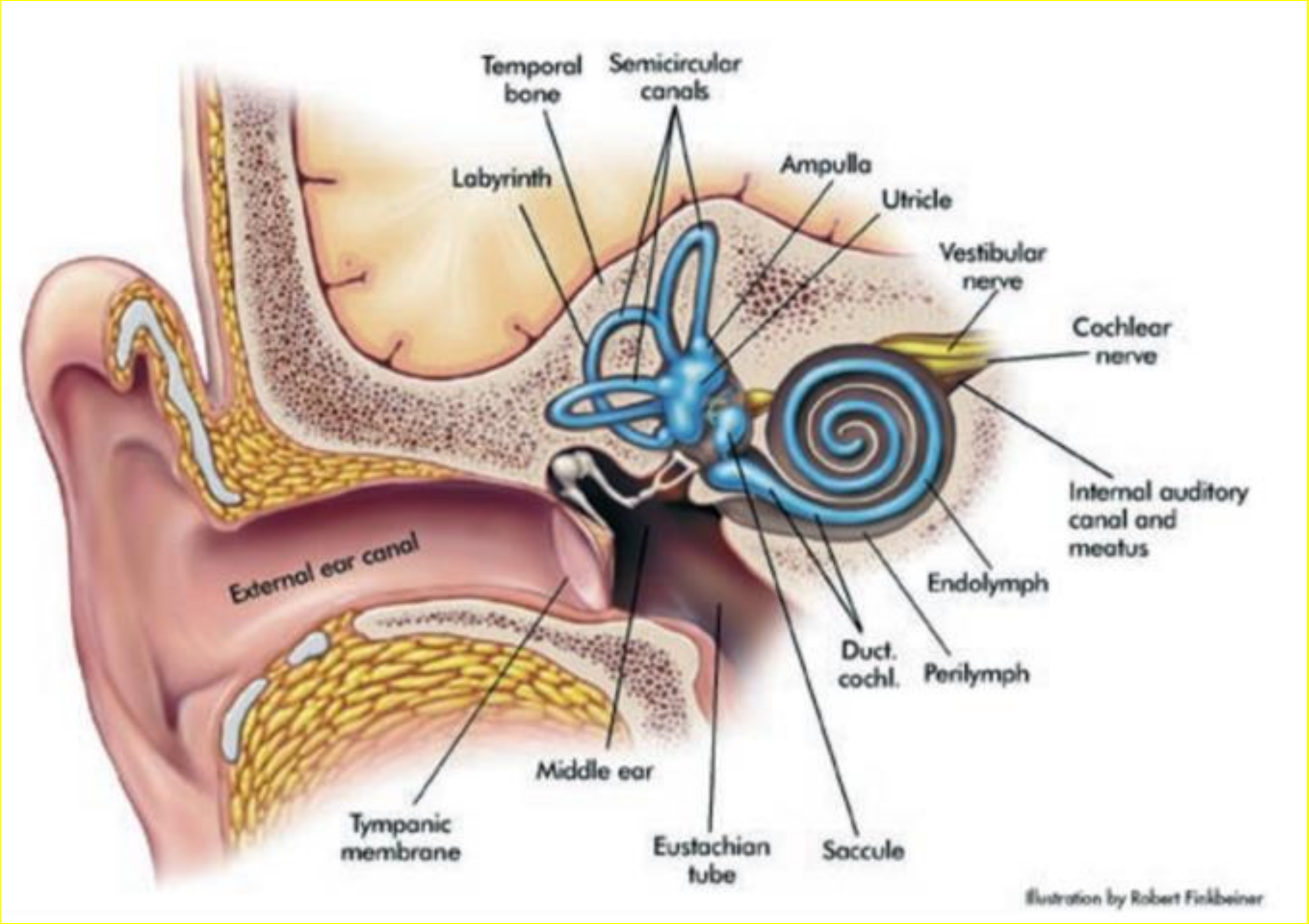
Vestibular system helps to maintain posture, balance, and eye fixation as we move in the environment. The vestibular system is pushed to the limit in our modern lifestyles.

Vestibular dysfunction generally presents with vertigo (sensation of world moving when it is not) and postural instability.

Clinical signs include nystagmus (repetitive, uncontrolled eye movements) and positive Romberg tests.

Vestibular disorders can be peripheral or central.

# *Anatomy of the Peripheral Vestibular System*





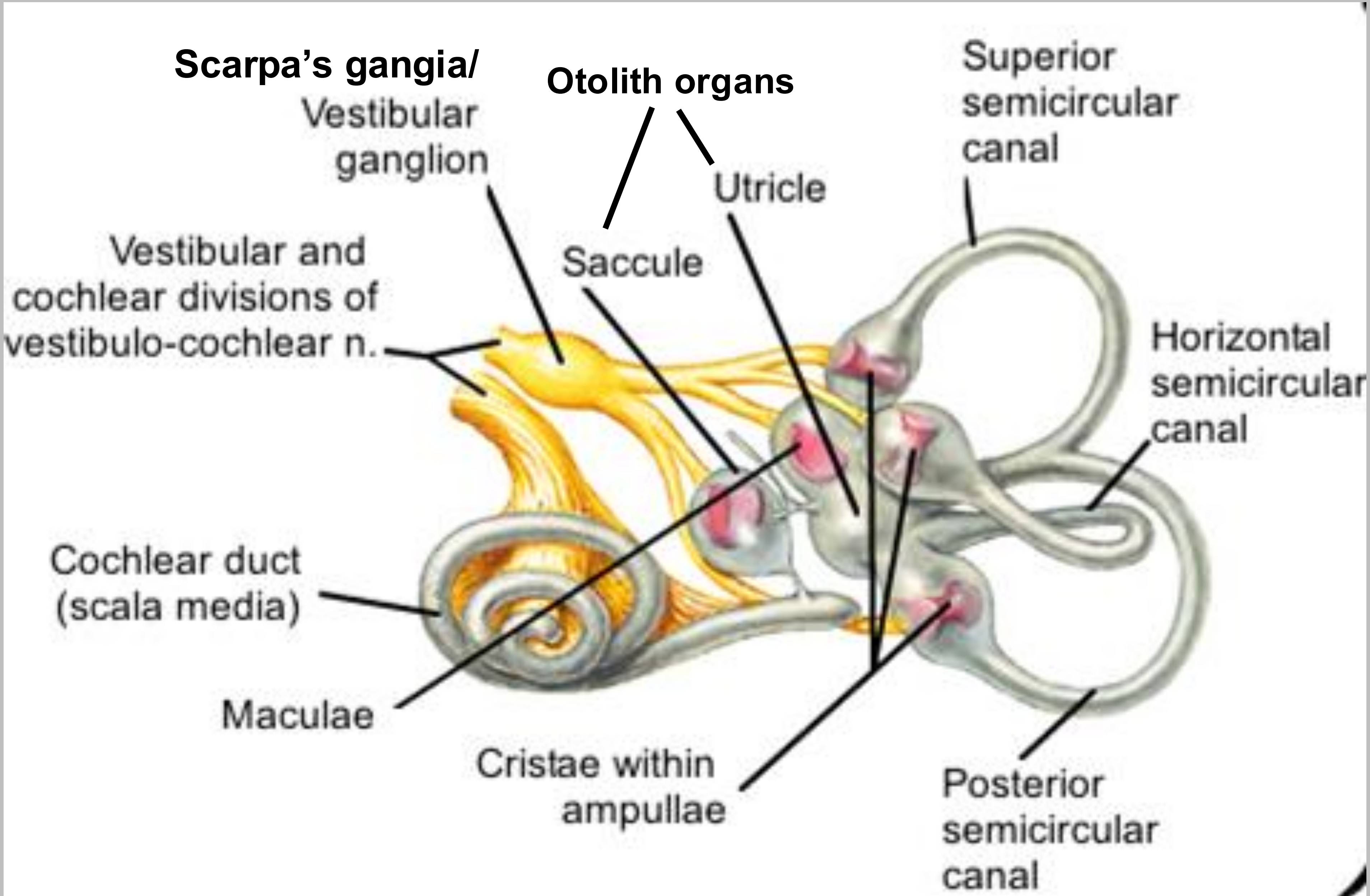
Otolith organs
- sacule
- utricle

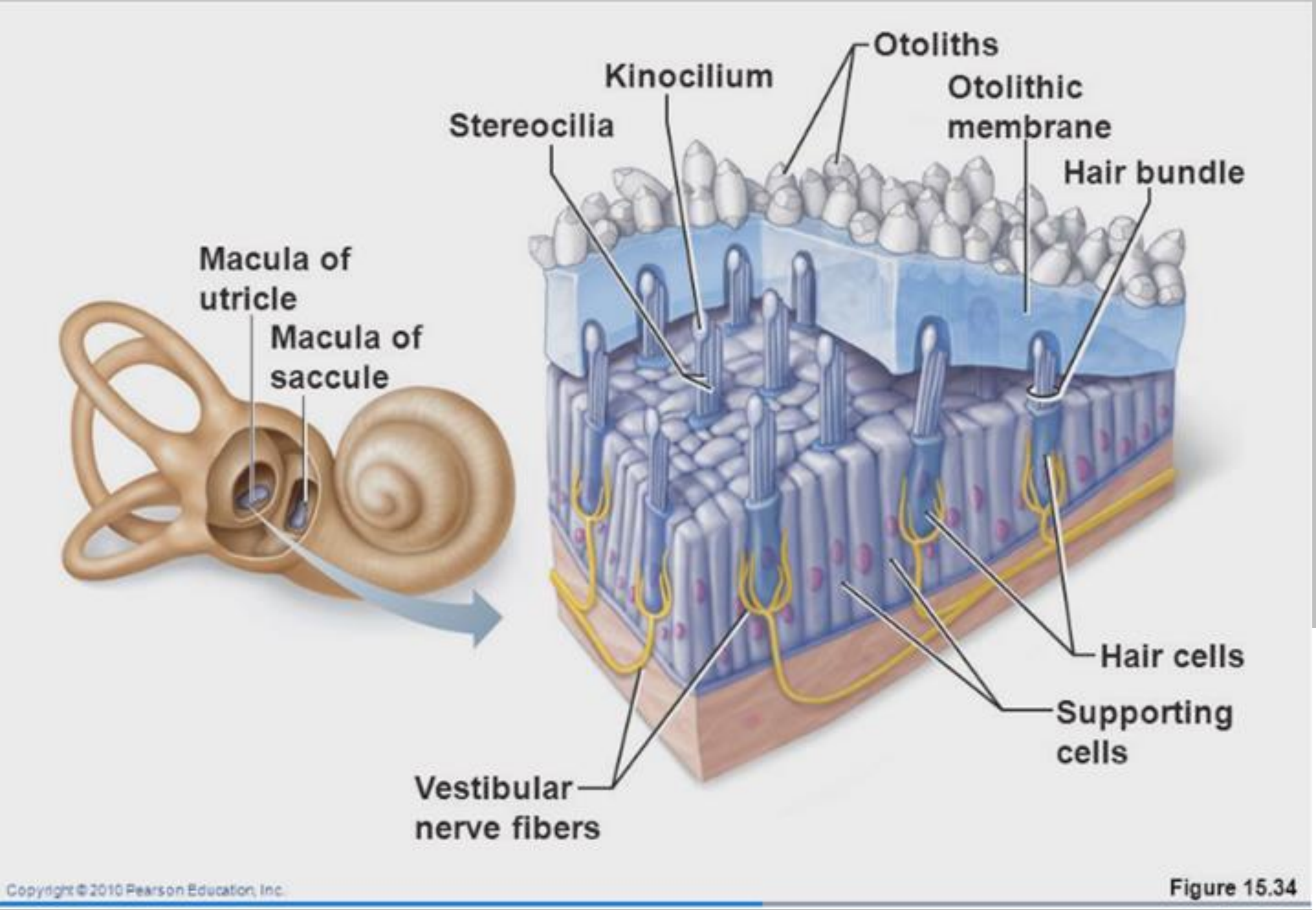
Semicircular canals

Endolymph vs Perilymph

Vestibular portion of CNVIII

# *Anatomy of the Peripheral Vestibular System*





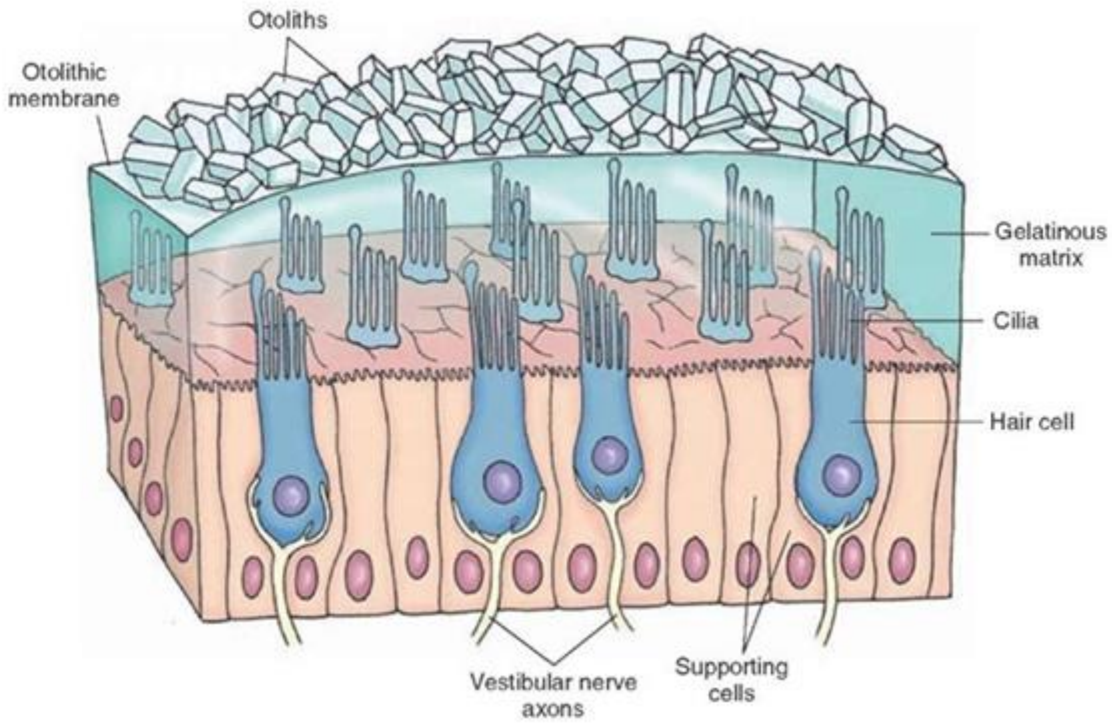


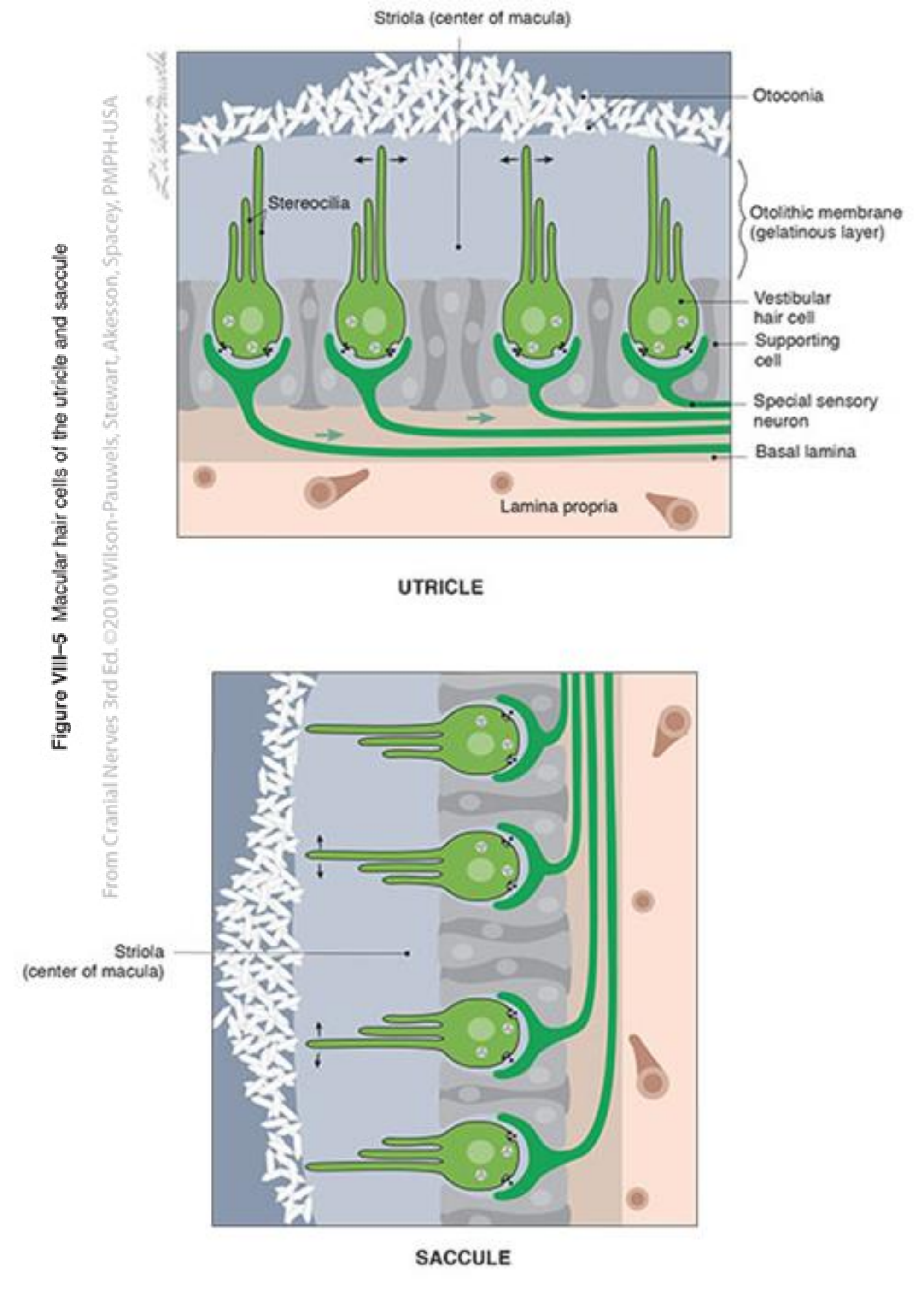
# Otolith organs

-Saccule & Utricle are cavities filled with endolymph

-Each has a macula lined with hair cells

-covered by a loose otolithic membrane which is high in specific gravity due to its gelatinous composition.

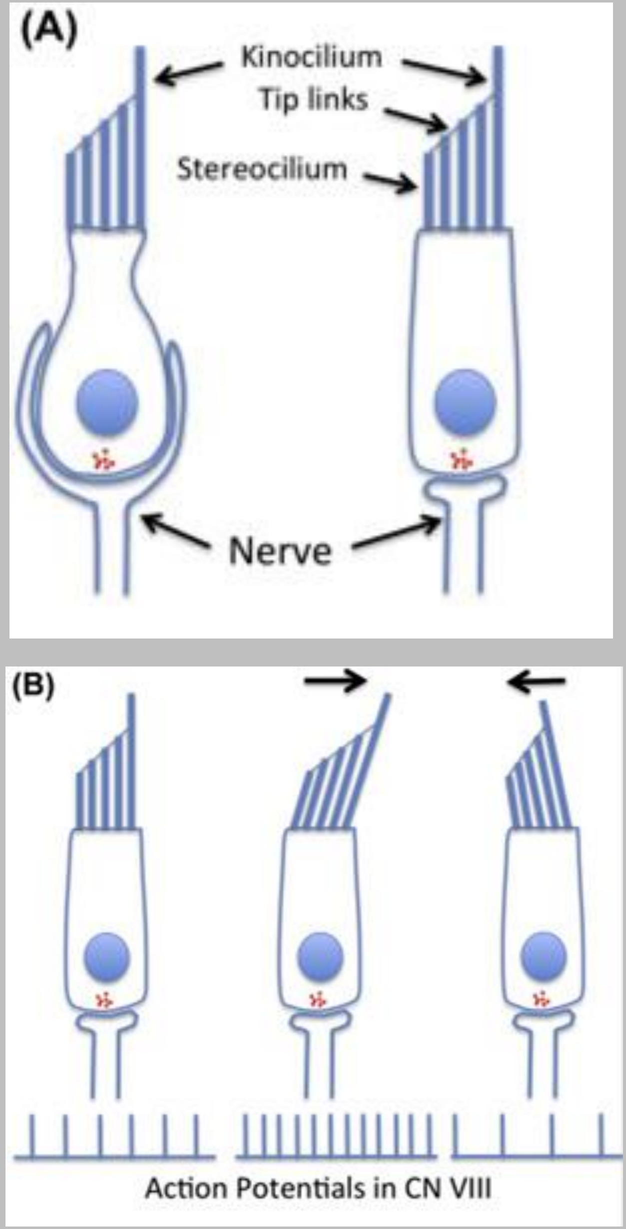


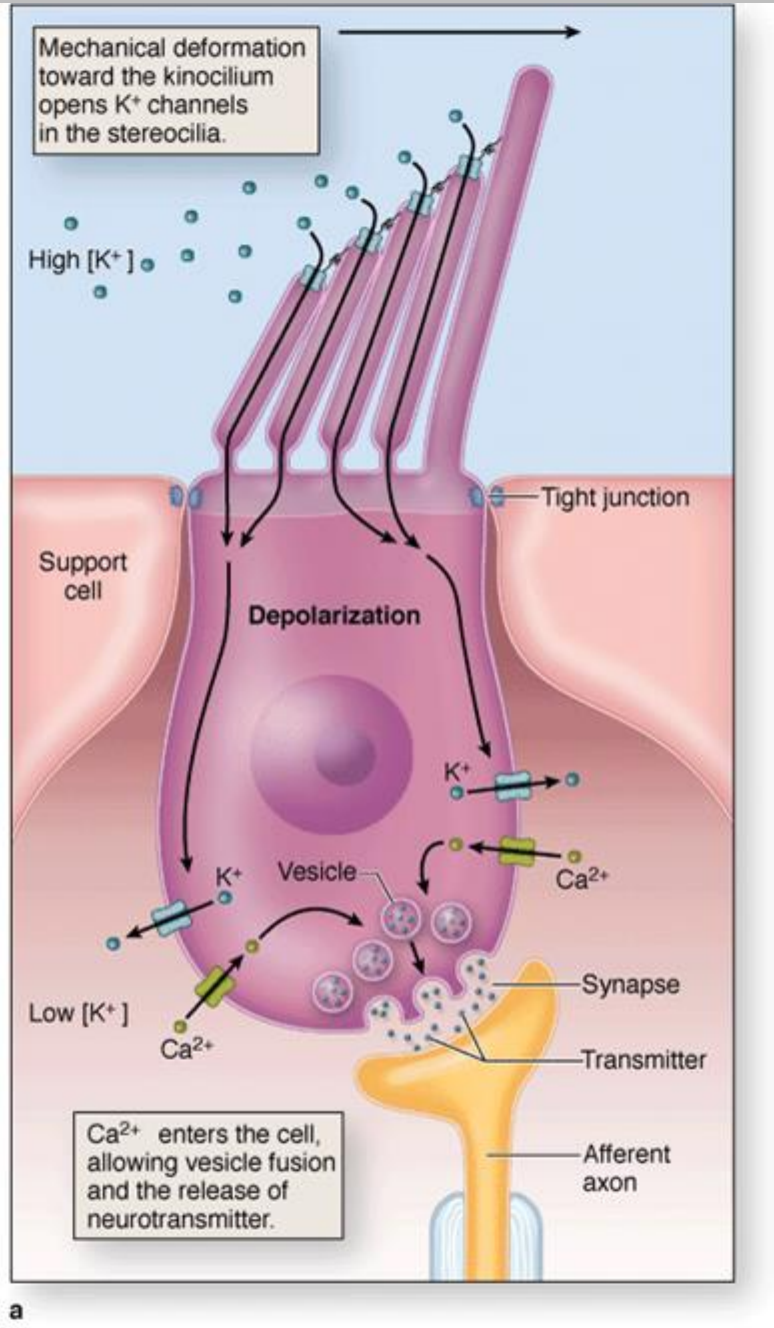



- Utricle – hair cells Upright
- Saccule – hair cells on Side

Encode the position and movement of the head with respect to gravity (linear acceleration) via activity of hair cells (deflection of cilia…tip links…K+/Ca+…glutamate…).

Bending in one direction causes depolarization. Bending in opposite direction causes hyperpolarization.






Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas, 12th Edition*: http://www.accessmedicine.com

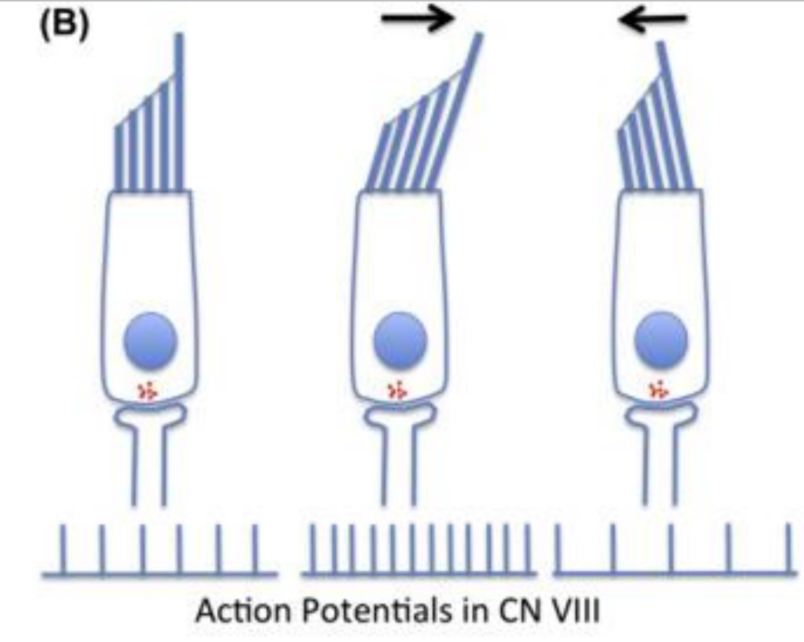


Deformation of hair cell cilia in the direction of the kinocilium results in depolarization of the hair cells via opening of K+ channels on tip links.

Depolarization causes voltage gated Ca++ channels to open resulting in transmitter fusion, exocytosis, and transmitter release.

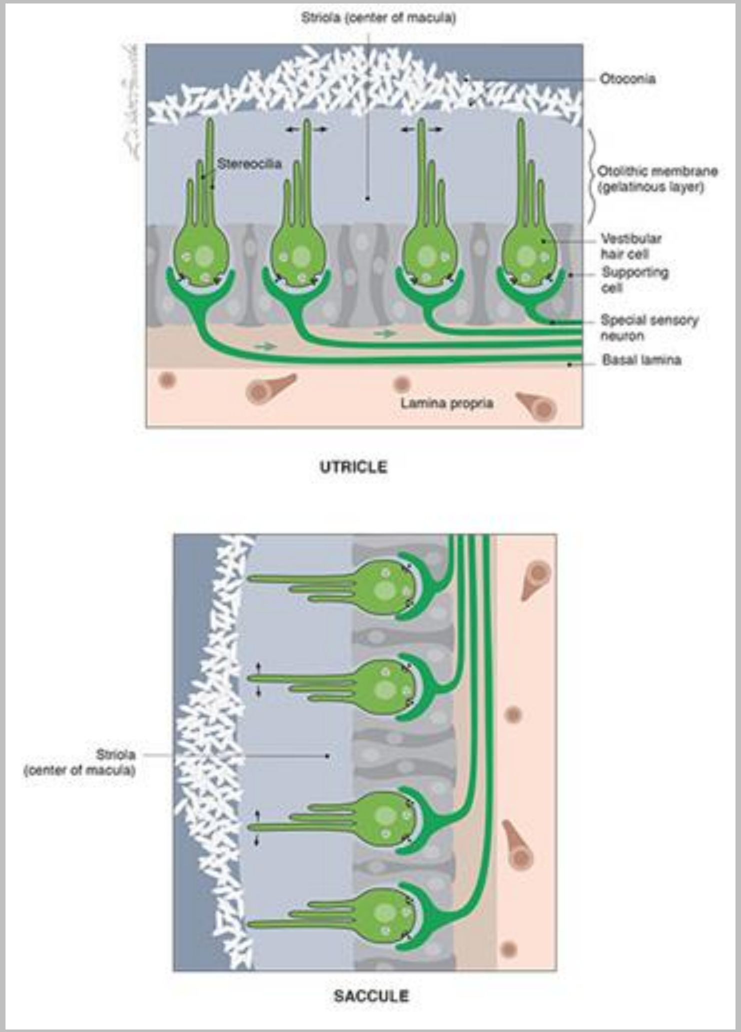
Binding of glutamate onto receptors on CNVIII cells causes Na+ influx and depolarizes these cells causing action potential generation.

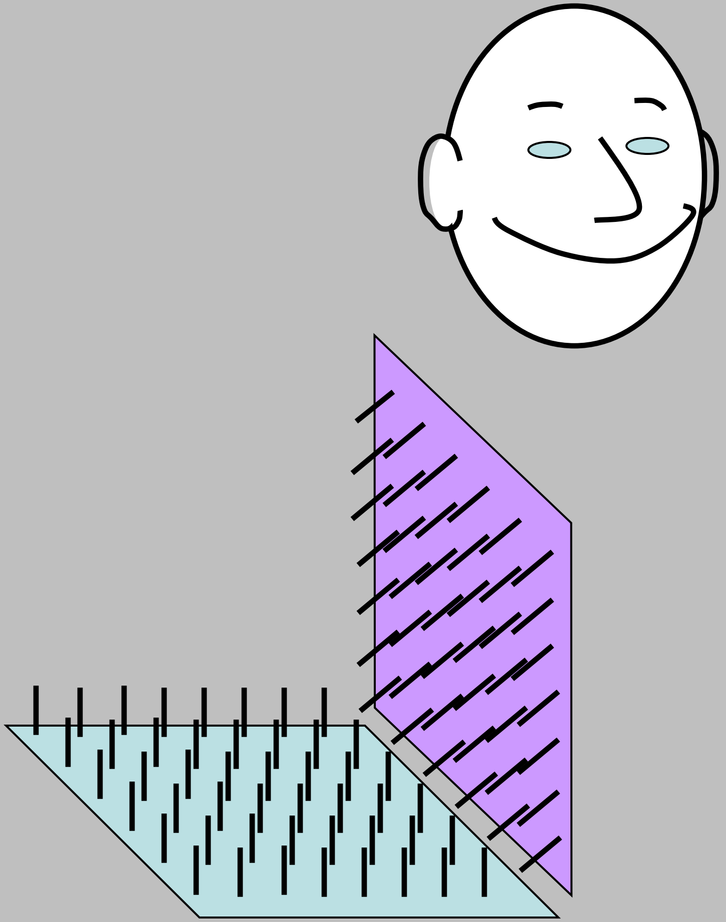
Deformation of hair cell cilia in the direction opposite of the kinocilium results in hyperpolarization.



# *Otolith organs: sensory transduction*

The **OTOLITH organs** are sensitive to gravity and **linear** head motion



# *Anatomy of the Semi-circular canals*

3 ducts emerging from the utricle each oriented along the 3 major planes (x, y, z)

Detects **angular head** movements along the 3 major planes (x, y, z)

Each contain an ampula where sensory cells are located.

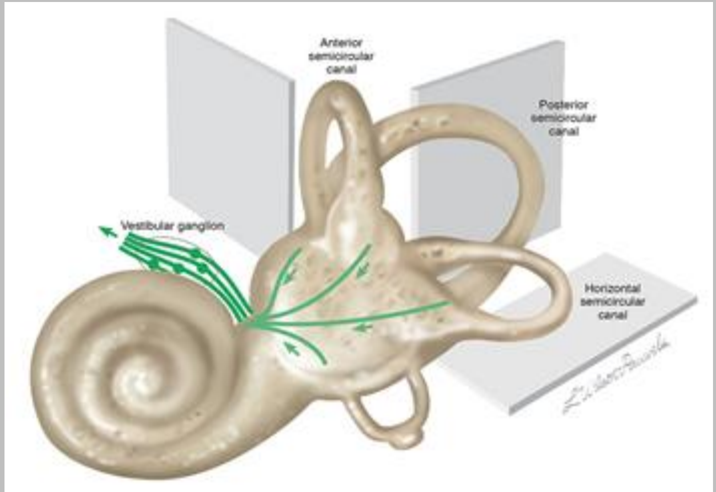
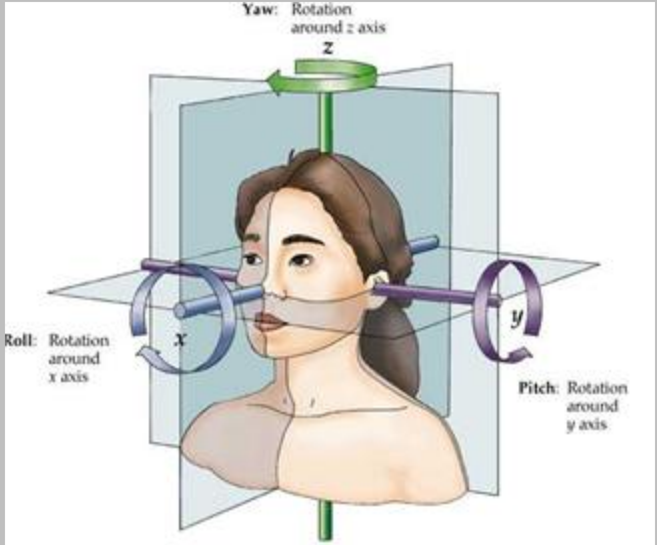



**Figure VIII–6** Planes of the semicircular canals. Note that they are at right angles to each other.

From Cranial Nerves 3rd Ed. ©2010 Wilson-Pauwels, Stewart, Akesson, Spacey, PMPH-USA



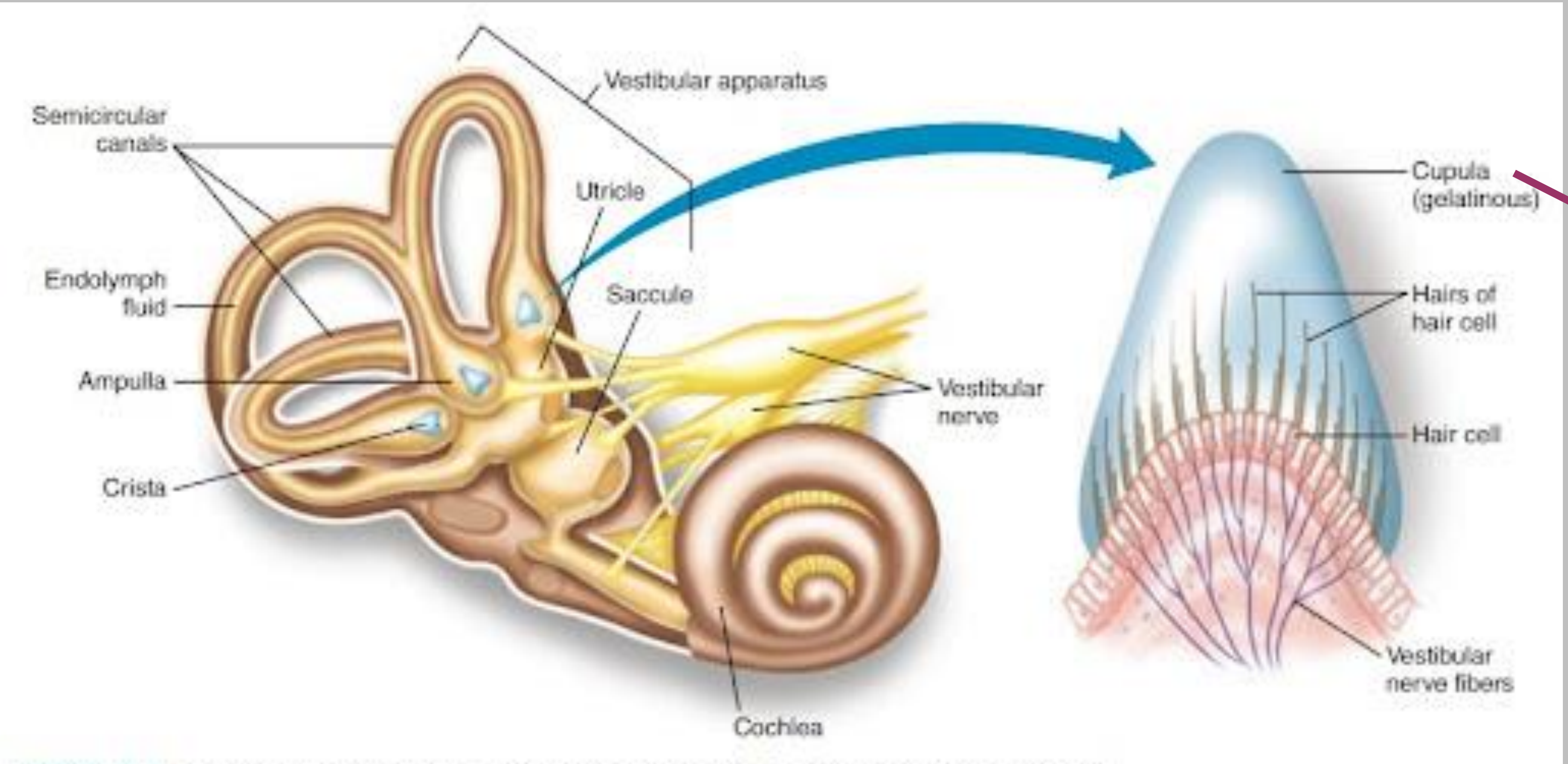
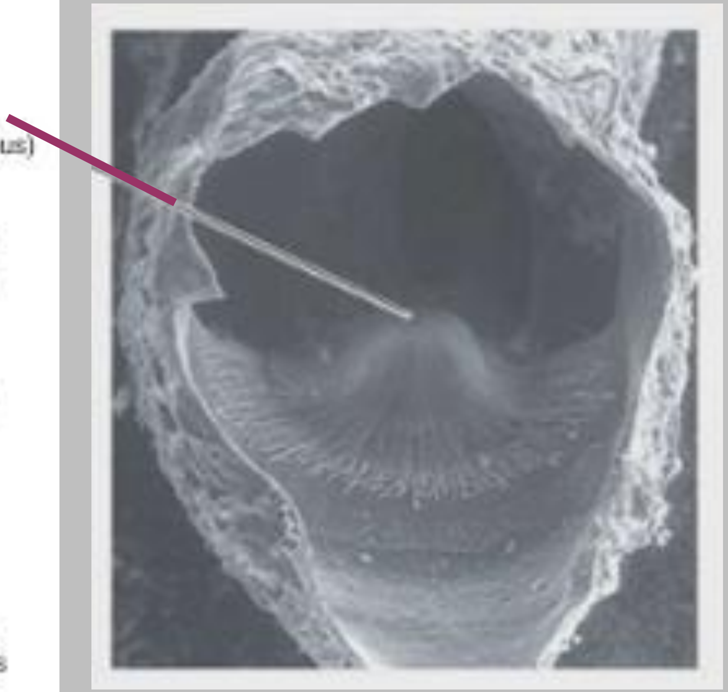
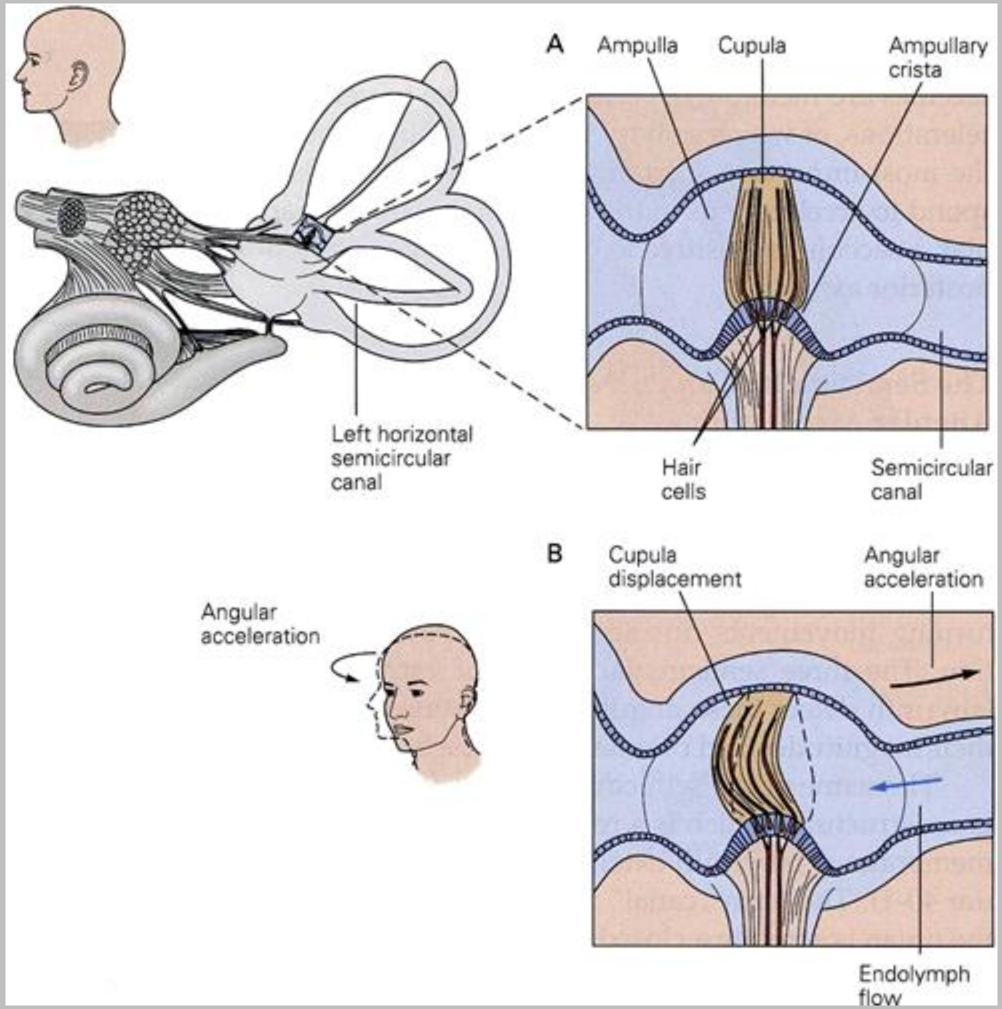
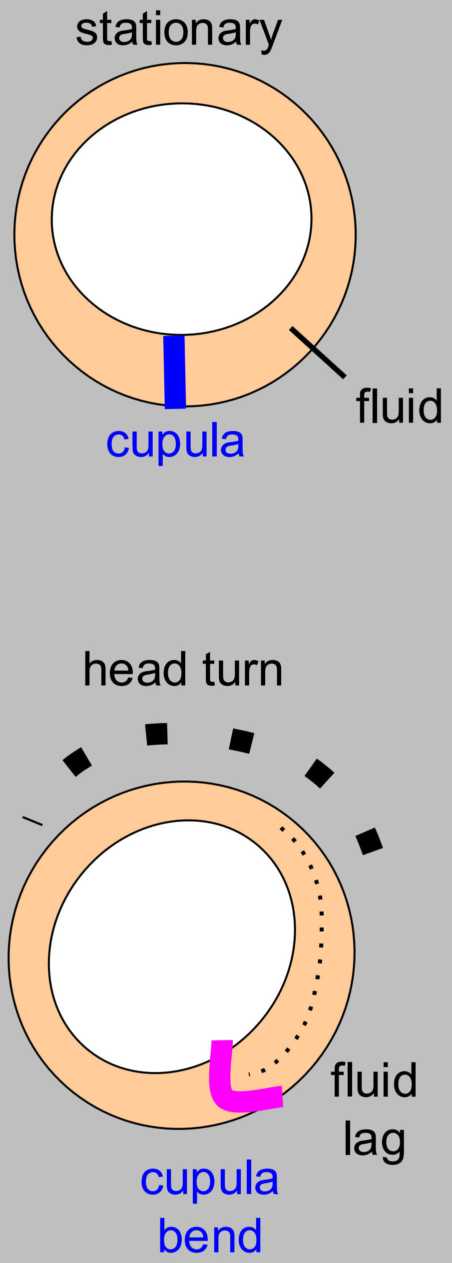




**FIGURE 15-7** The crista ampullaris is located within the ampulla of each semicircular canal.



# *Semicircular canals: sensory transduction--* **angular** head motion



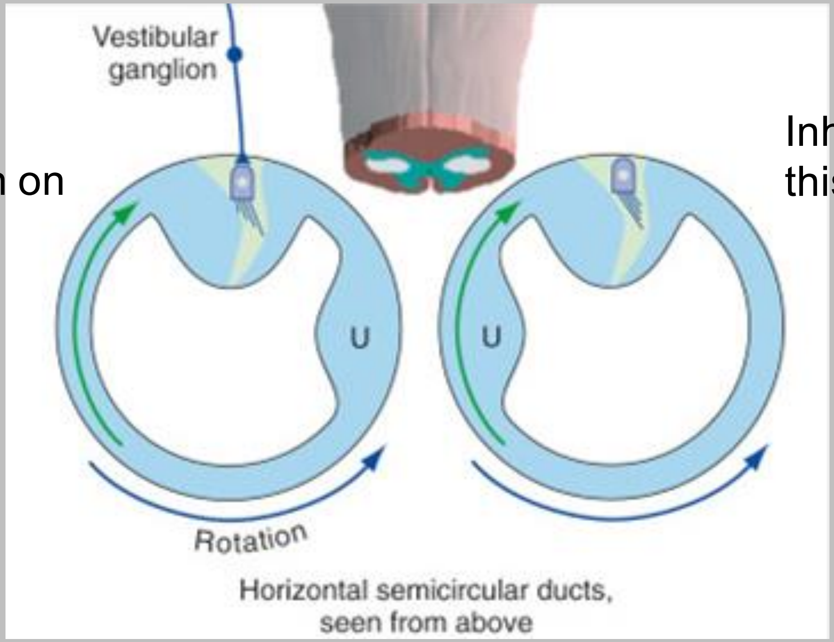




The cristae respond to rotary acceleration

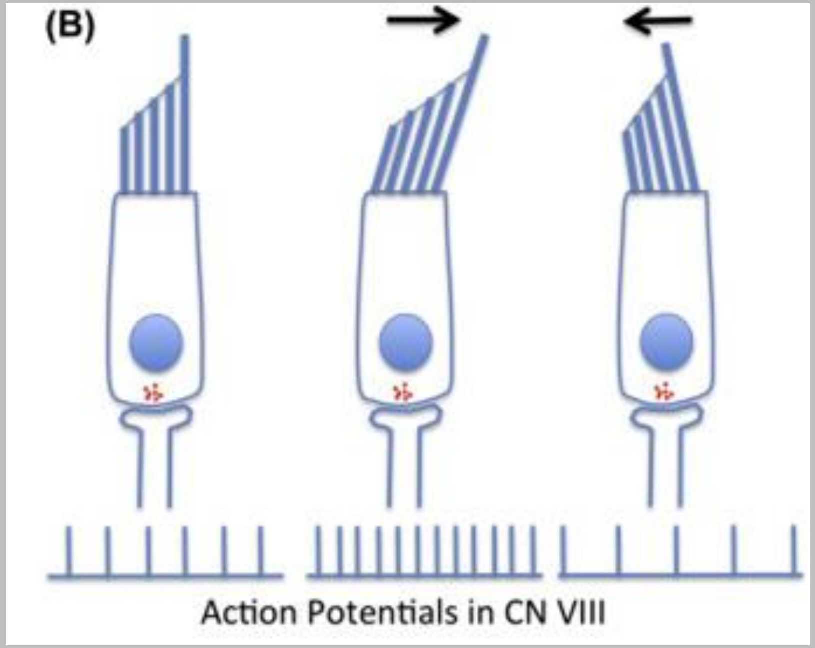
During a rotation, the endolymph lags behind the head acceleration which causes deflection of the cupola opposite to the head movement

Hair cells are deflected causing a change in their membrane potential causing more or less transmitter release onto CNVIII cells.

Excitation on this side




Inhibition on this side




The cristae respond to rotary acceleration

Endolymph lags the head acceleration deflecting the cupola opposite to the head movement

Hairs and cilia in cupula bend

The largest cilia are arranged toward the midline

Excitation occurs when the flow causes bending of cilia toward midline.

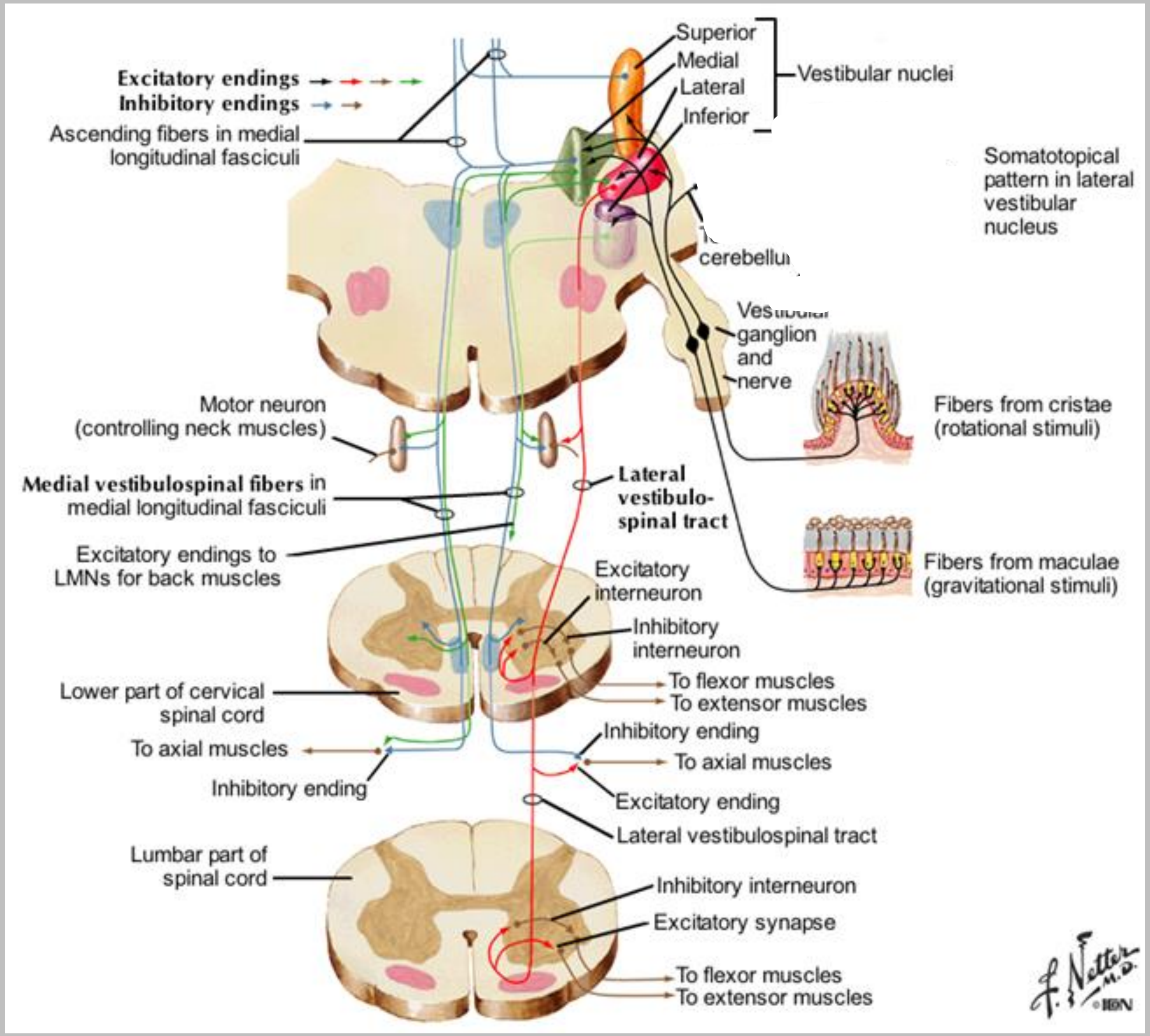
Inhibition occurs when the flow causes bending of cilia toward lateral

The hair cells synapse to the peripheral process of cells in the vestibular ganglion

# Vestibular pathways:

Vestibulo-spinal pathways controls corrective actions during movement to maintain posture.

- Lateral vestibulo-spinal tract
    - **Lateral and inferior vestibular nuclei ipsilateral projection.** Project onto anterior horn and innervate proximal muscles of limb **contraction of forelimb and hindlimb muscles** to maintain balance and upright posture
- Medial vestibulo-spinal tract
    - **Medial vestibular nucleus** project **bilaterally** (medial longitudinal fasciculus) innervate neck muscles including SCM, **stabilization of head position**



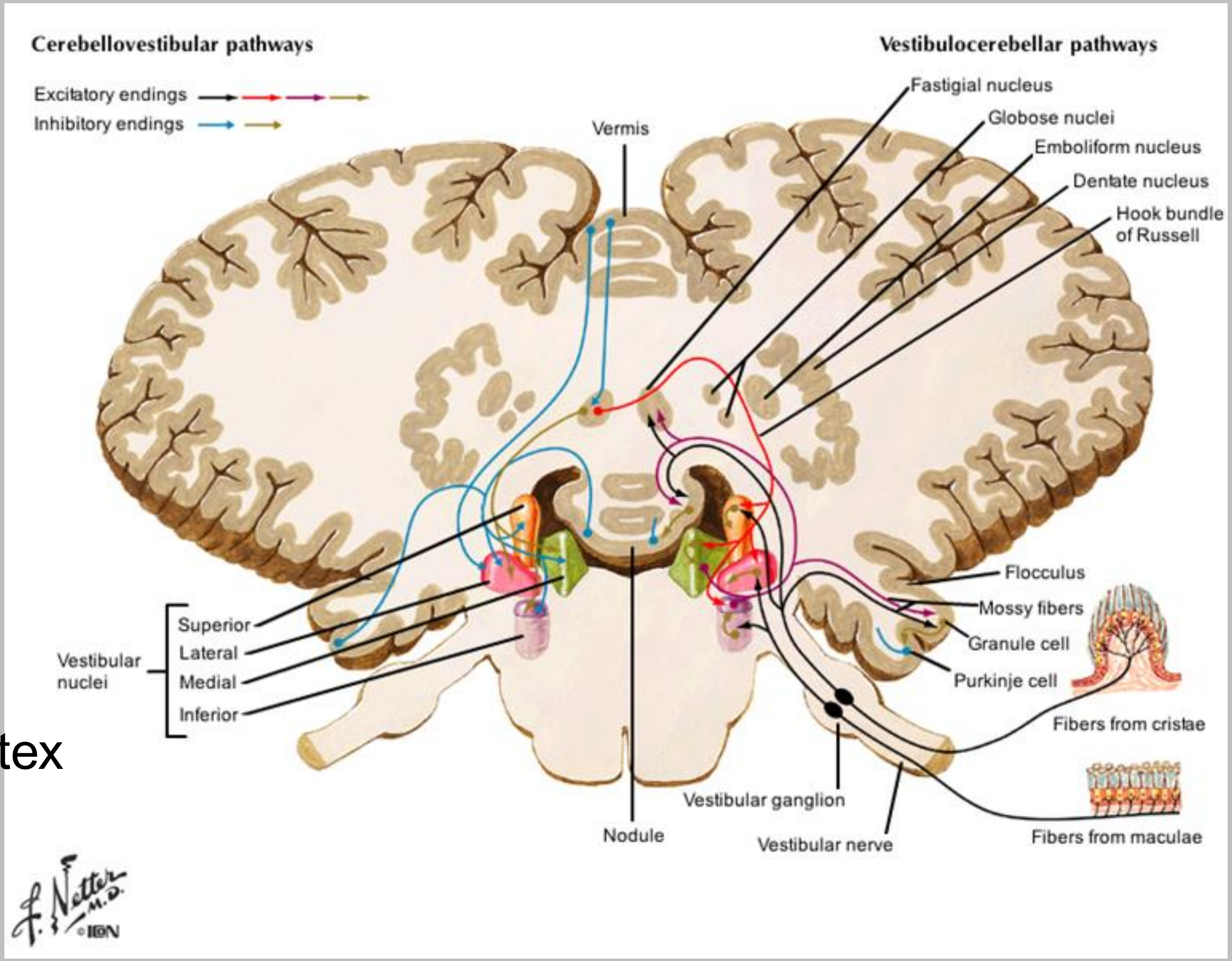
# Vestibular pathways:

## 2] Vestibulo-cerebellar pathway

Controls corrective actions during movement to maintain posture.

Includes projections to deep cerebellar nuclei & cerebellar cortex (flocculus & nodule)

Includes only Ipsi projections

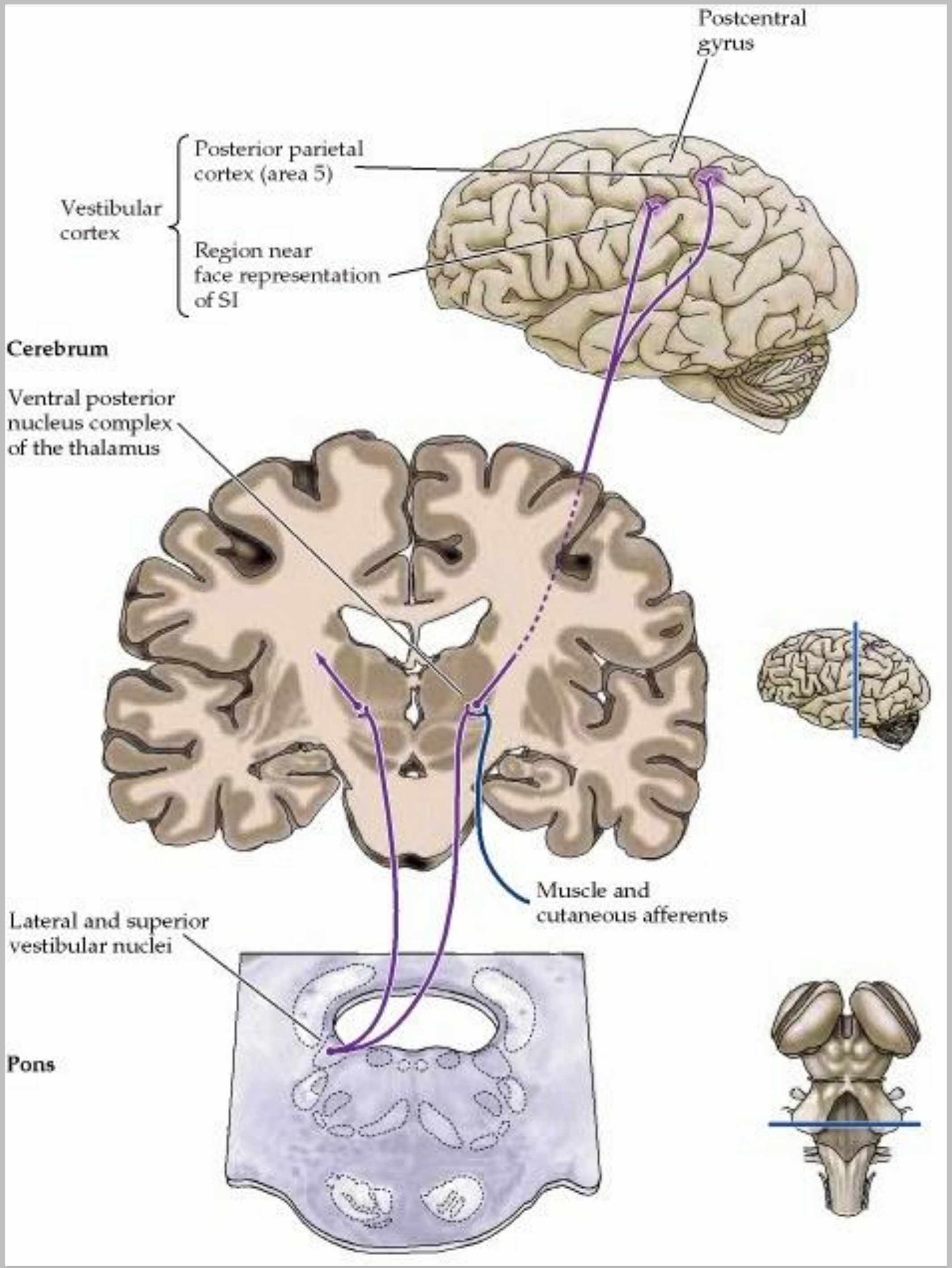


## Vestibular pathways:

### 3] Vestibulo-thalamic pathway

Essential for conscious perception of body and eye position/movement in relation to environment

Includes Ipsi & Contra projections

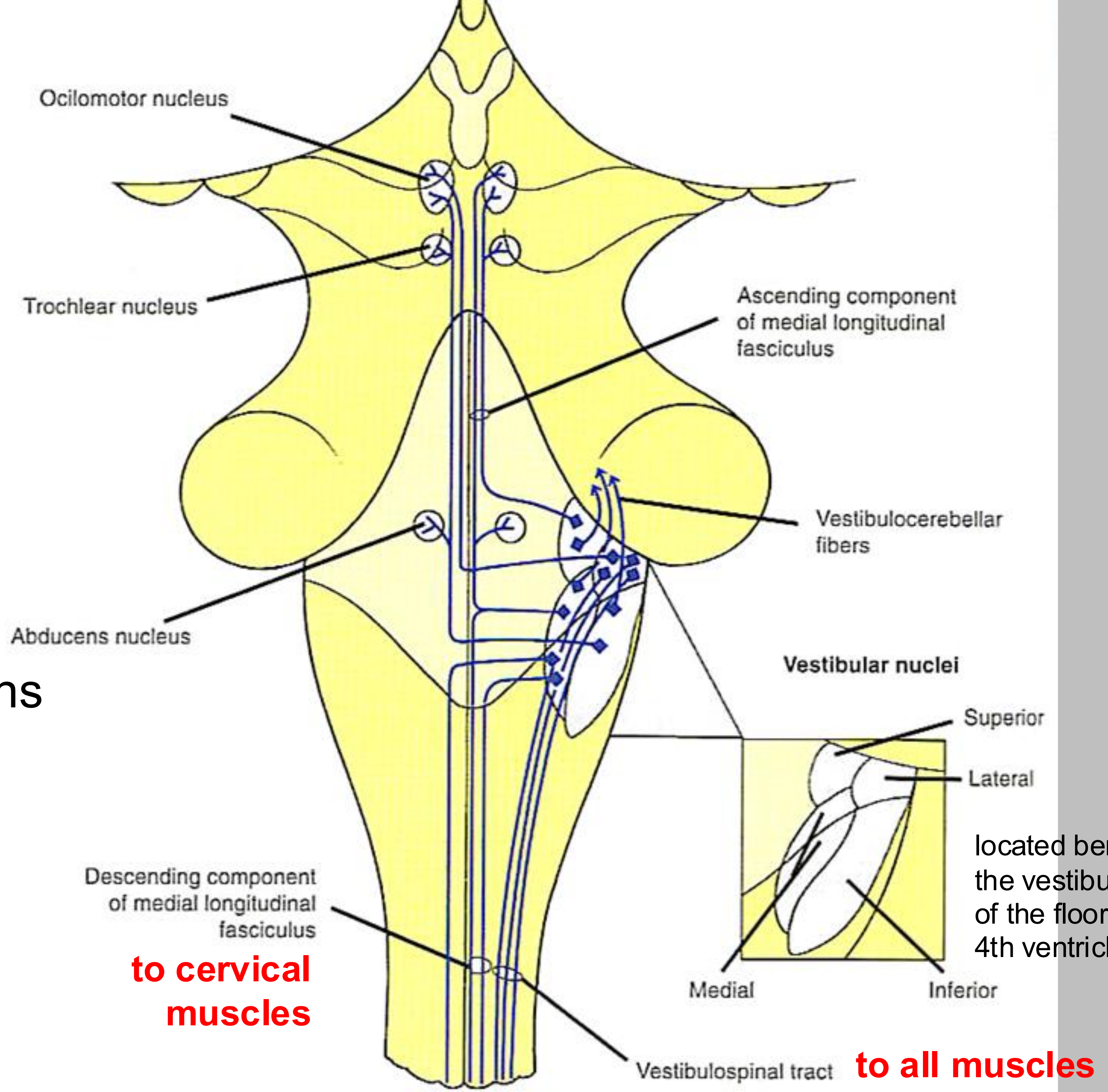


Vestibular pathways:

4] Vestibulo-oculo pathway

Controls corrective eye movements to maintain posture and fixation during movement.

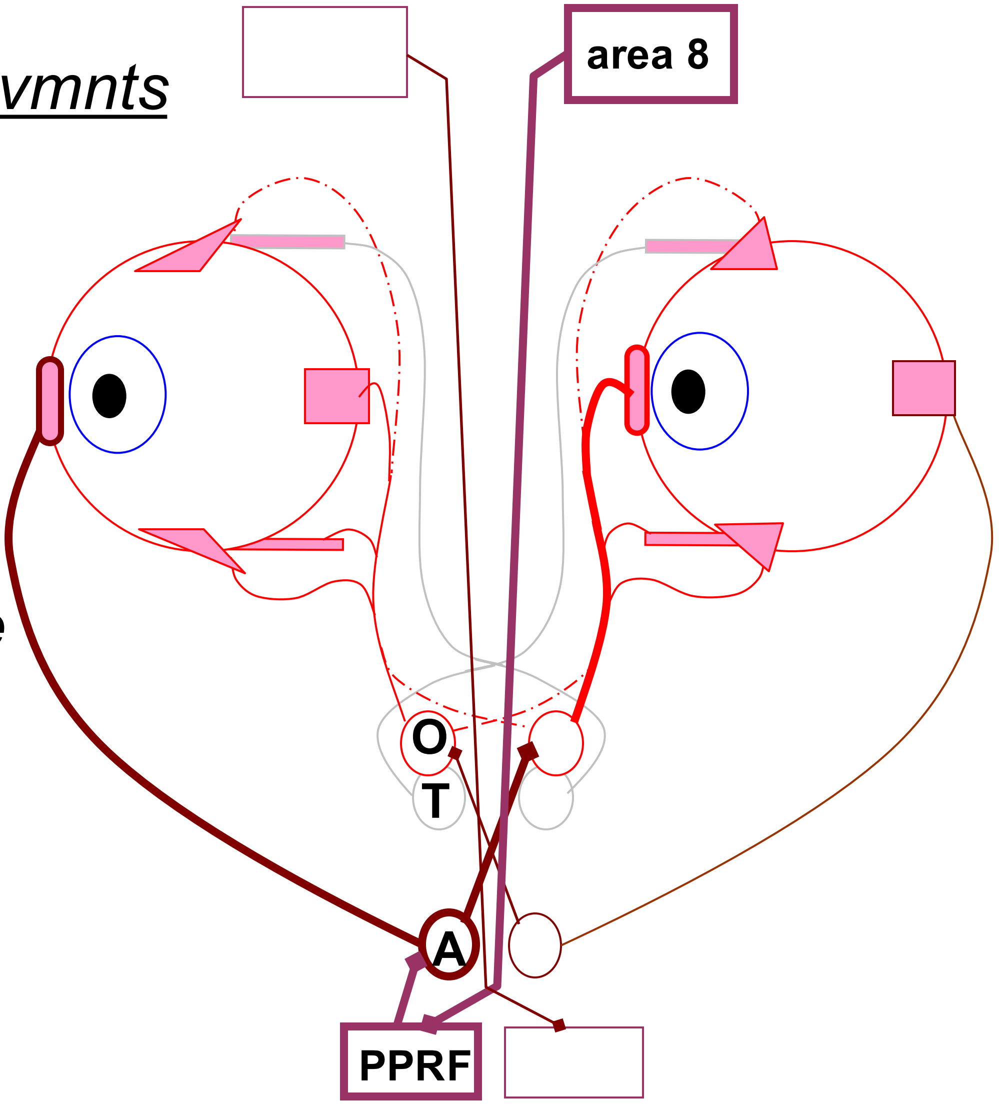
Includes Ipsi & Contra projections

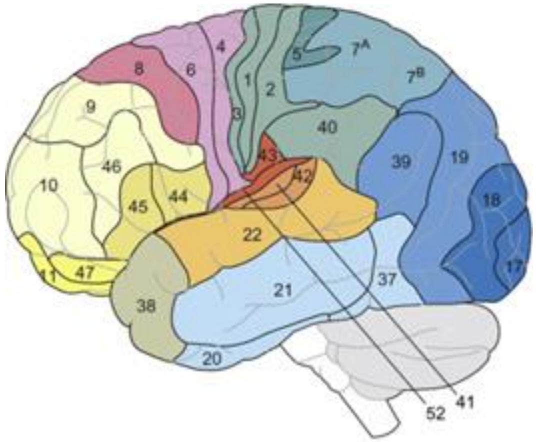


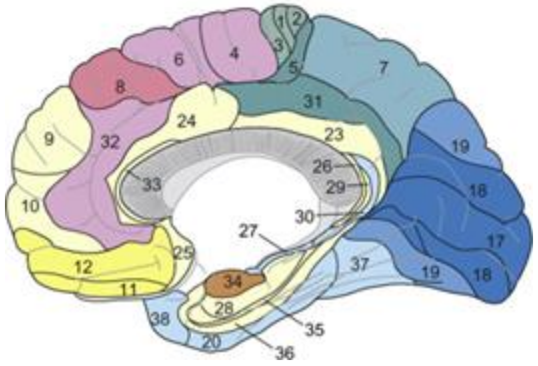

located beneath the vestibular area of the floor of the 4th ventricle

# ***Voluntary** lateral eye mvmnts*

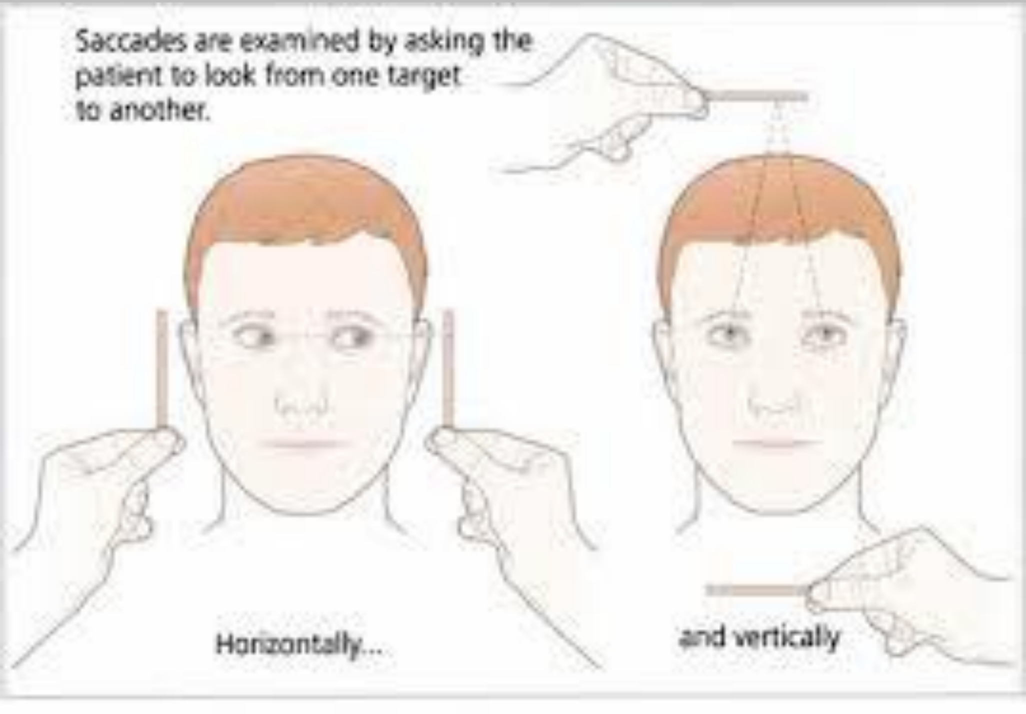
*originates in*
*cortical **area 8**,*
*the frontal eye field*
*and is mediated by*
*the **P**aramedian **P**ontine*
***R**eticular **F**ormation*
*(PPRF)*



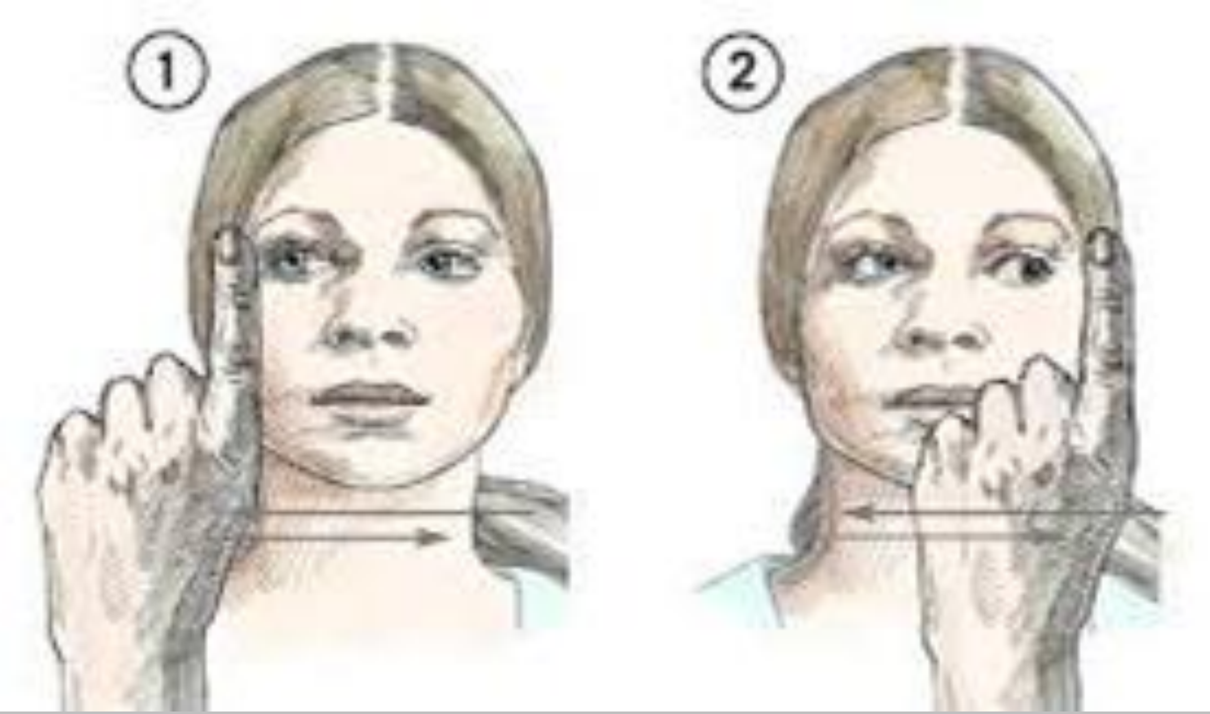





Clinical testing of **Saccadic** eye movements



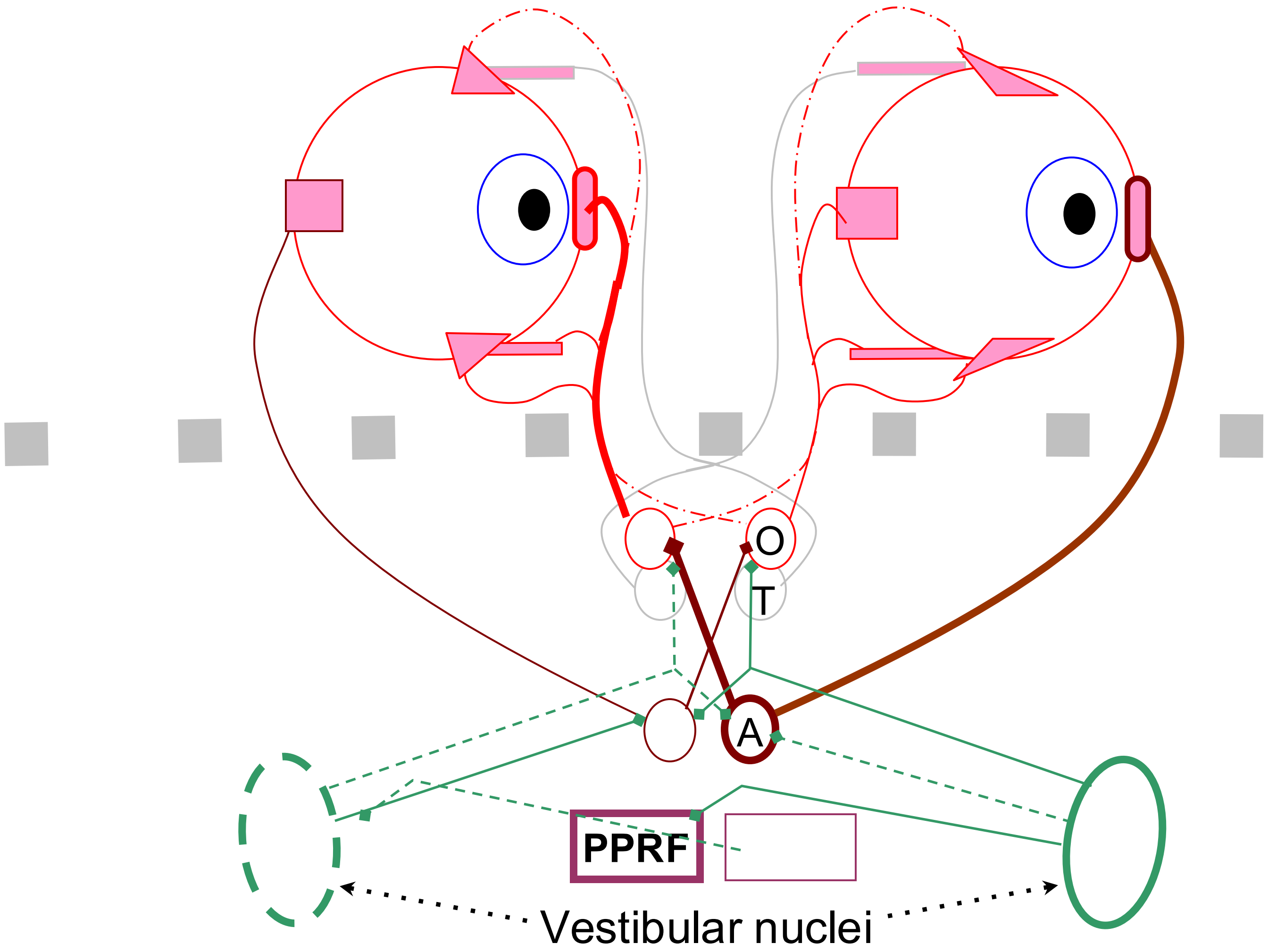

Clinical testing of **Smooth pursuit** eye movements

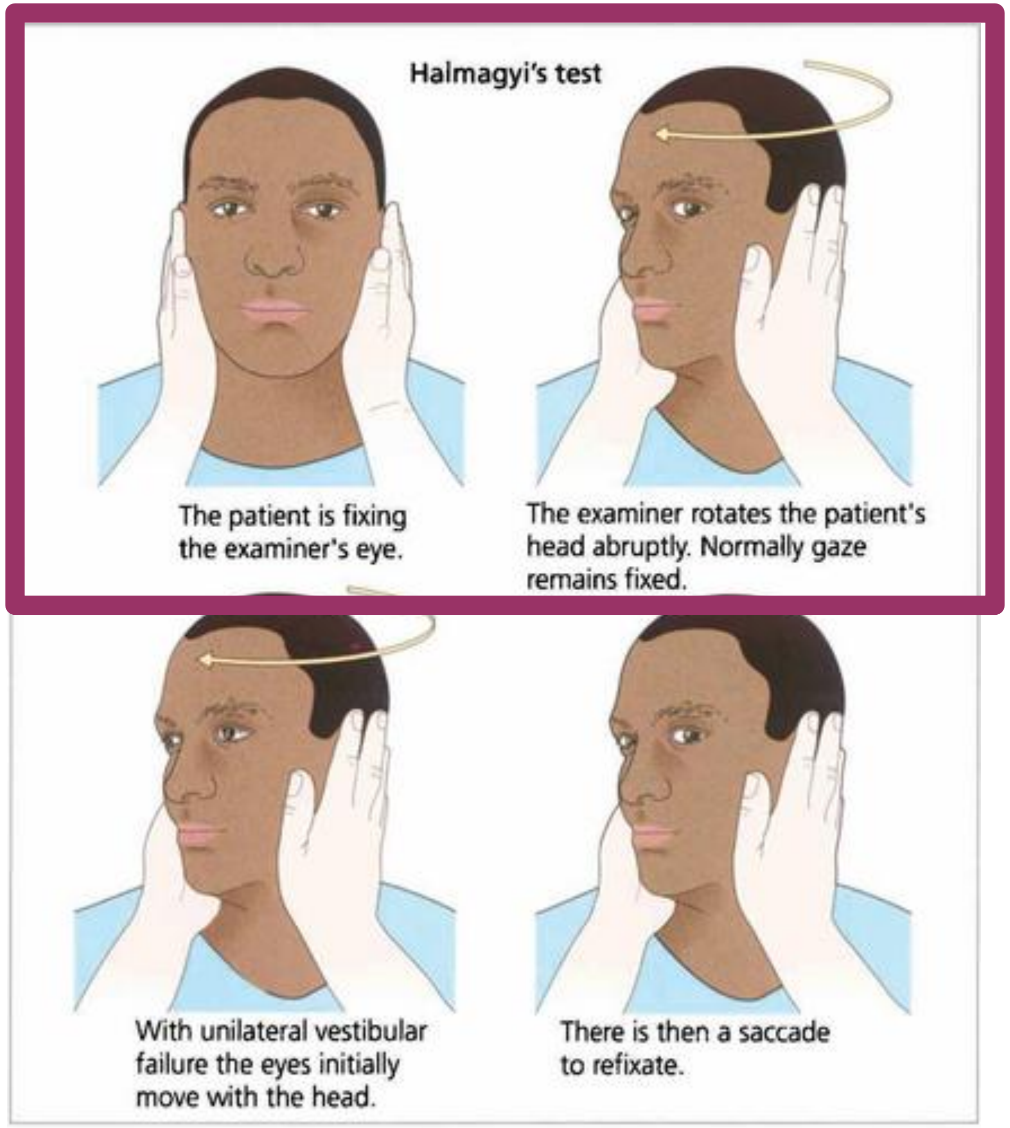



These eye movements are voluntary and do not engage the vestibular system when there is no movement of the head or body

# VOR- Vestbulo Occular Reflex

Maintaining eye fixation on a visual target even when head moves. Eyes move opposite head.



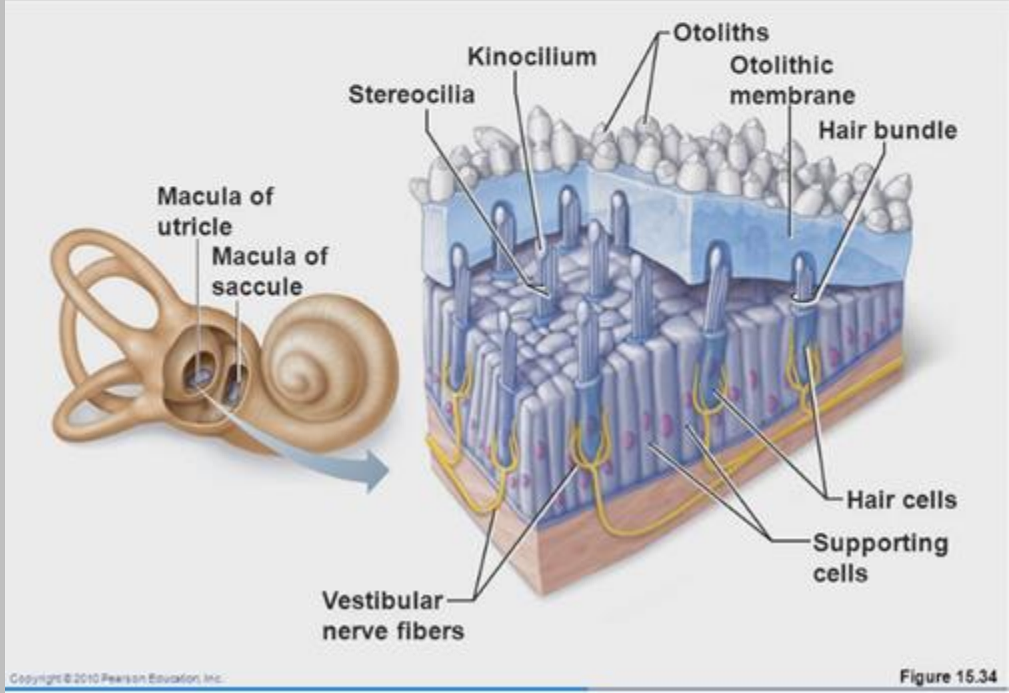



# *Benign paroxysmal positional vertigo - BPPV*

benign = does not worsen over time or reflect additional pathology
paroxysmal = abates relatively quickly (< 1 min)
positional = is elicited with particular head movements
vertigo = rotary vertigo (compelling spinning sensation) includes nystagmus and nausea

Most cases are idiopathic  can be linked to head trauma or viral infection. Also, more common in older individuals and certain occupations (e.g. automechanics and plumbers).

Associated with loose otoliths that get stuck in semi-circular canals and disturb normal movement of endolymph and cristae.



# *BPPV testing in patients that present with vertigo: Dix-Hallpike maneuver*

*Nystagmus seen during the test = positive sign & indicative of peripheral vestibular dysfunction such as loose otoliths*

If test is *negative*, vertigo is likely related to dysfunction of vestibular circuits in central nervous system


Video-oculography to record eye movements

doi/full/10.1056/NEJMicm0907386

Video at http://www.dizziness-and-balance.com

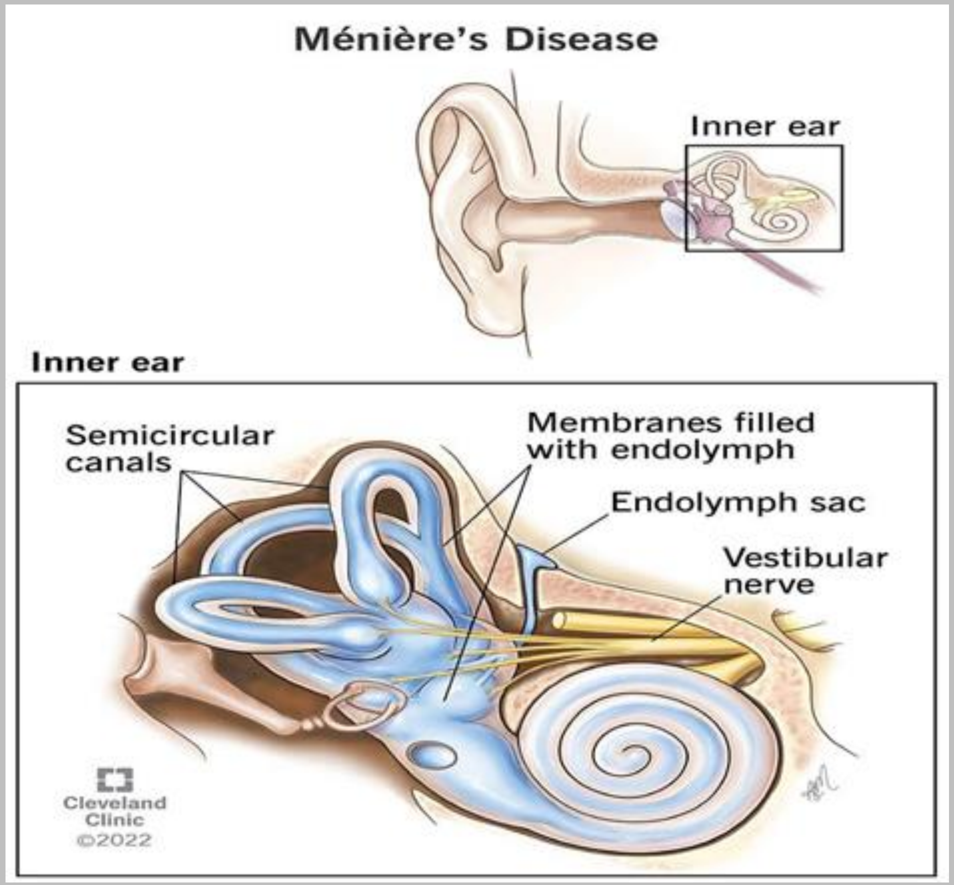
# *Benign paroxysmal positional vertigo treatment: Epley manuever*

The Epley manuever attemps to return the otoconia to the utricle.

80-90% initial success rate, 50% recurrence


# Meniere Disease (PV)

- A cause of peripheral vertigo ( presents worse that central vertigo)
- Caused by improper endolymph resorption in the vestibule
- Etiology: Idiopathic, but can be due to viral infections, autoimmunity. Also some genetic or environmental factors involved.
- Symptoms: Recurrent episodes of peripheral vertigo, fluctuating unilateral sensorineural hearing loss and unilateral tinnitus (Meniere triad).
- Treatment: antihistamines, benzodiazepines, anti-emetics.

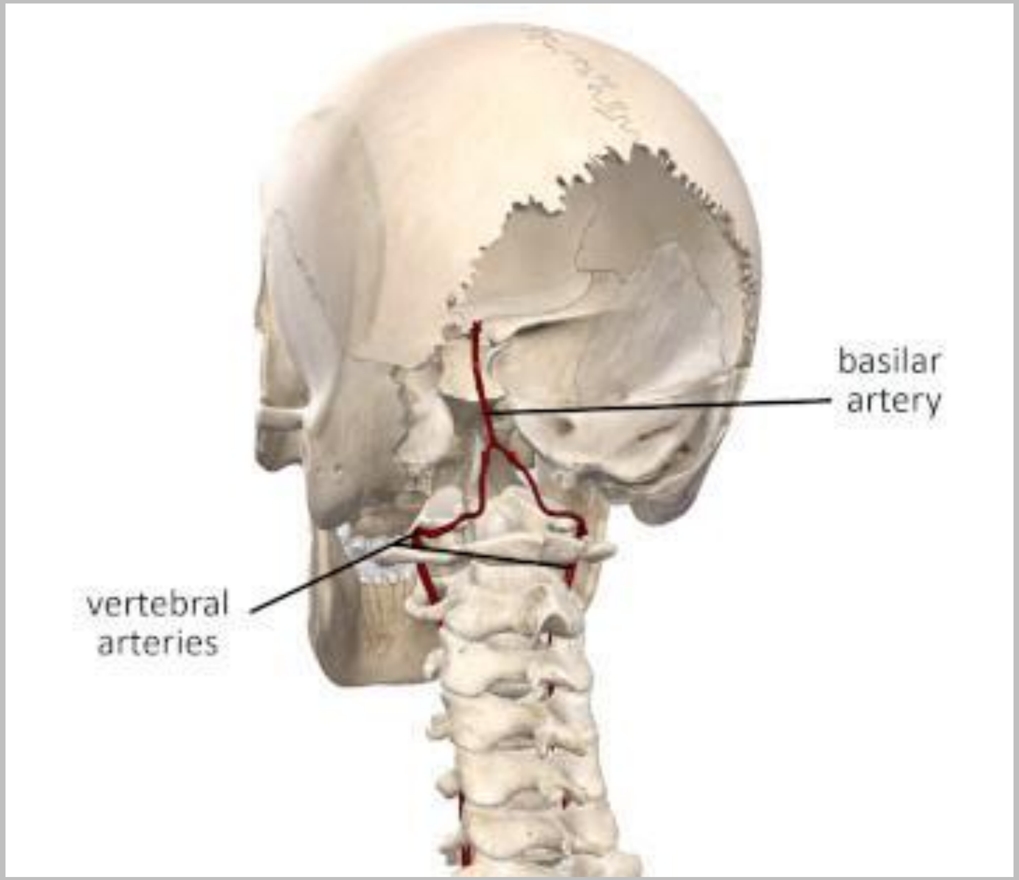


**Labyrinthitis**

- Caused by **Inflammation of the membranous labyrinth** of the inner ear.
  - Membranous labyrinth has endolymph which has lots of K+ in it.
- Etiology: bacterial or viral infection, and or manifestation of systemic autoimmune disease. HIV can also cause it.
- Sx: severe vertigo, N/V, tinnitus, and/ or hearing loss.
- Usually pt makes full recovery, but some are left with residual balance or hearing issues.
- Labyrinthitis is an INFECTION of inner ear
- Meniere's is not necessarily an infection, but can be caused by infection. Meniere’s is also more likely to present with multiple recurrent episodes, unlike labyrinthitis.

# Vertebrobasilar Insufficiency

- Caused by inadequate blood flow through the posterior circulation. The medulla, cerebellum, pons, midbrain, thalamus and occipital cortex can all be affected in VBI.  2 vertebral arteries cross midline at C6 transverse process to become basilar artery.
- It may occur as a TIA and usually causes by atherosclerosis of vertebral arties or basilar artery. It can also be caused by traveling embolism.
- Risk factors include: HTN, smoking, older age, male, positive family history, obesity, diabetes and hyperlipidemia
- Symptoms include: Central vertigo, syncope, nausea, visual or auditory disturbances, dysphagia and hemiparesis.

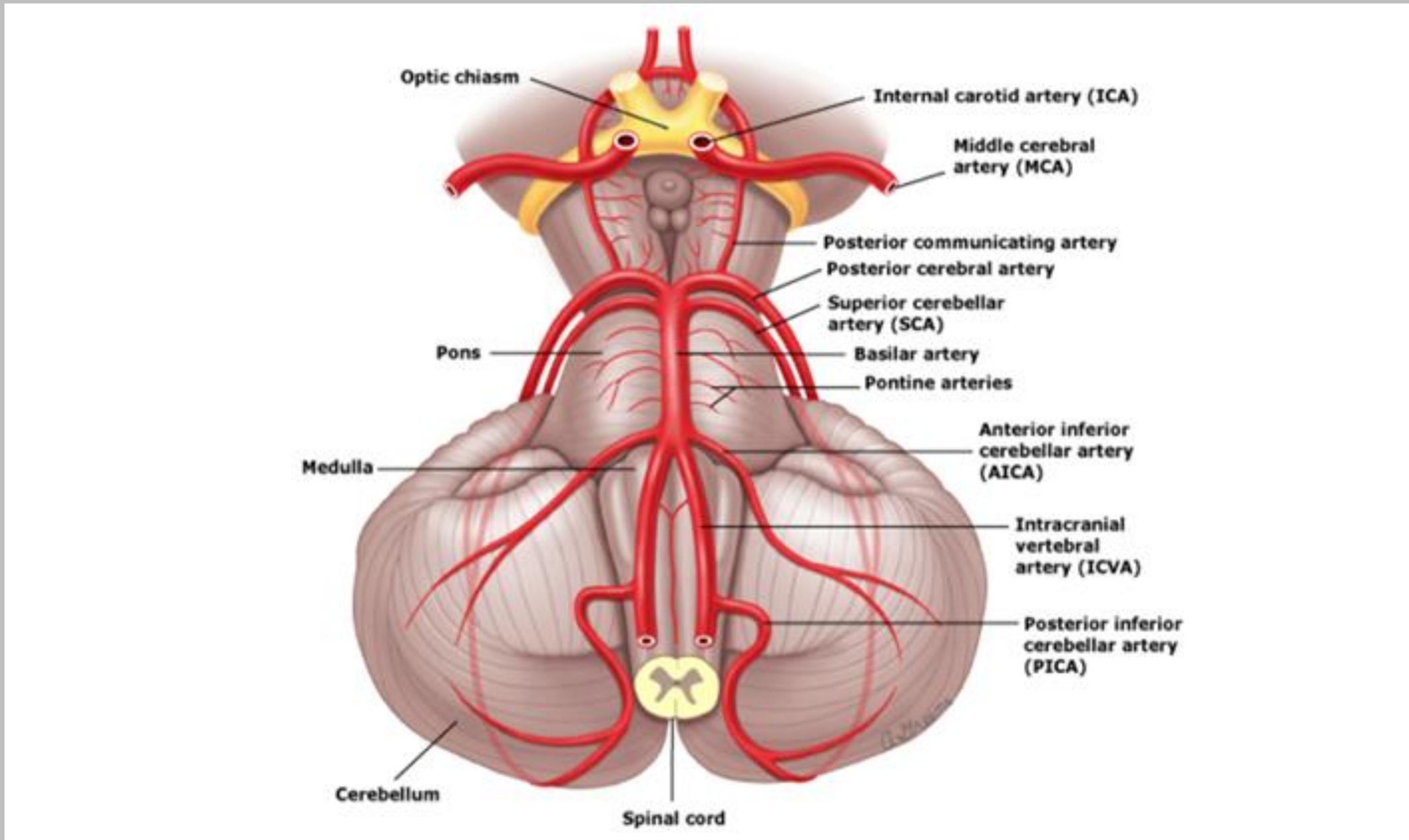


# Ototoxic drugs

- Aminoglycosides
- Furosemide
- Cisplatin
- Quinine
- Aspirin/salicylates
- Erythromycin
- Vancomycin
- Opioids
    - no need to memorize this yet

# Posterior Circulation Stroke

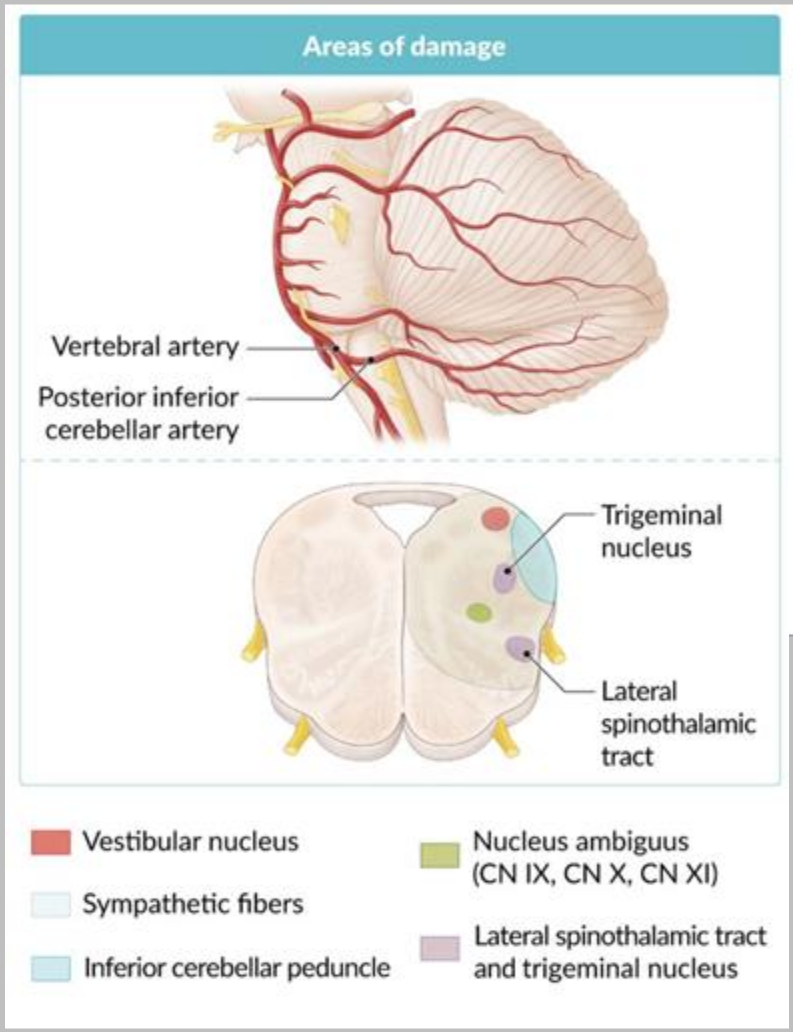
- PCA strokes account for 20% of all strokes, most PCA strokes are ischemic→ atherosclerosis, dissection or embolism
- Most commonly occur in: Posterior cerebral artery (PCA)>posterior inferior cerebellar artery (PICA)> superior cerebellar artery (SCA)>anterior inferior cerebellar artery (AICA).
- Pt can present with central vertigo from vestibular nuclear involvement at pontomedullary junction
- Ataxia (cerebellar involvement)
- Nystagmus
- Vomiting
- Headache
- Motor weakness

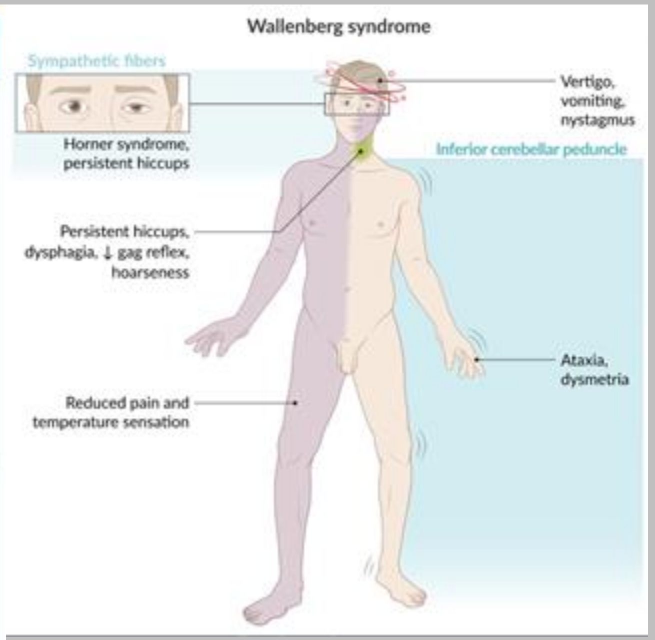


# Lateral Medullary /Wallenberg Syndrome (CV)

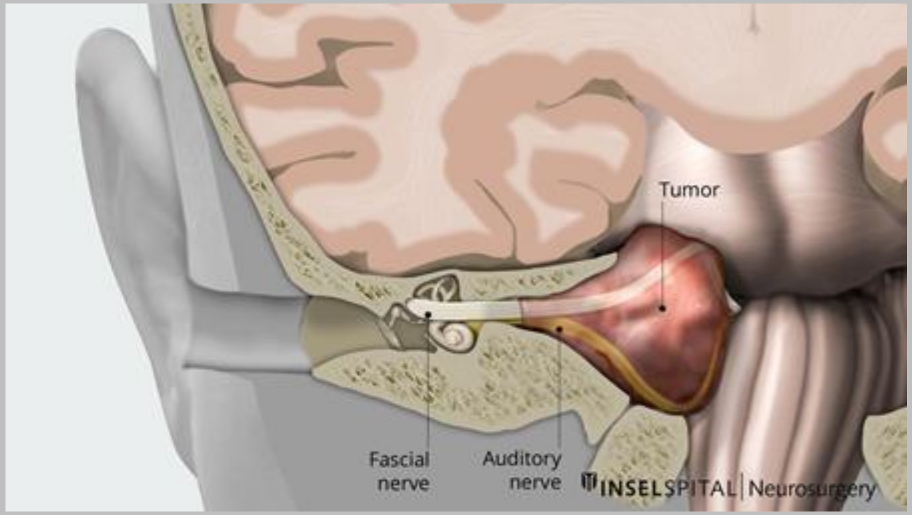
- A cause of central vertigo.
- Caused by occlusion of PICA.
- Causes an Ipsilateral
  - vertigo and nystagmus (central, beats in direction of gaze) Vestibular nucleus damage
  - ataxia and dysmetria from inferior cerebellar peduncle insult, carry signal from bipolar cells from vestibular ganglion to cerebellum and brainstem.
  - Dysphagia, hoarseness and dysphonia from nucleus ambiguus insult
  - Horner Syndrome and hiccups from insulting sympathetic fibers in area.

- Contralateral:
  - loss of pain and temp in trunks and limb (LSTT)



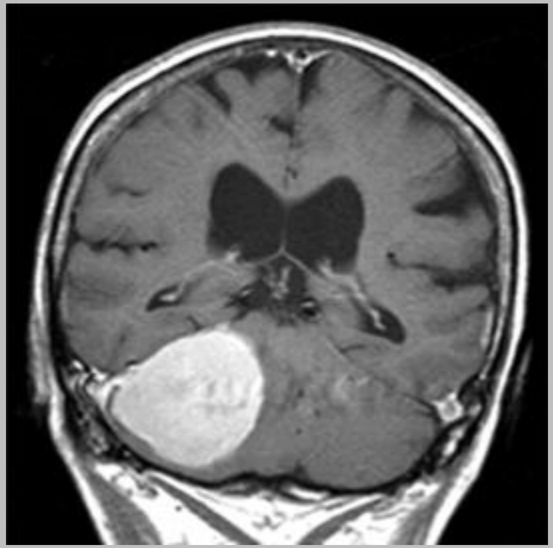



# Posterior Fossa Tumors

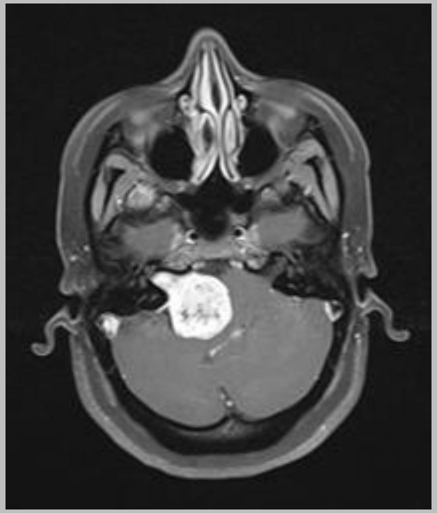



- Vestibular Schwannoma, (CV/PV)
  - Benign tumor that arises from the Schwann cells of the CN VIII.
  - Often also involves CN VII due to proximity: bells palsy. CN 7 leaves through stylomastoid foramen and CN8 and Cn7 run very close together. So a VS can also cause CNVII issues.
- Posterior fossa meningiomas
  - Brain tumors that arise from the meningeal layer.
  - Compression of vestibular nuclei, flocculonodular lobe, cerebellum and other vestibular pathway.

Right inferior tentorial meningioma



Vestibular Schwannoma w/cerebellar compression



# Central vs Peripheral Vertigo

| **Central Vertigo** | **Peripheral Vertigo** |
|---|---|
| Gradual onset | Sudden onset |
| Mild intensity | Severe intensity |
| Mild nausea/vomiting | Intense nausea/vomiting |
| Associated neurologic findings typically present | No associated neurologic findings |
| Mild nystagmus | Relatively intense nystagmus |
| Position change has mild effect | Position change has intense effect |
| No refractoriness—can repeat the “tilt” test and patient responds every time | Rapidly refractory—cannot repeat the “tilt” test; patient will not respond again |
| Patient falls to same side as lesion | Patient falls to same side as lesion |